**D 32717** **(Pages : 3)** **Name.....**

**Reg. No.....**

**FIRST SEMESTER M.Sc. (CBCSS) REGULAR/SUPPLEMENTARY DEGREE EXAMINATION, NOVEMBER 2022**

Mathematics

MTH1C02—LINEAR ALGEBRA

(2019 Admission onwards)

Time : Three Hours Maximum Weightage : 30

## Part A

*Answer* **all** *questions.*
*Each question carries* 1 *weightage.*

1. Let $\mathbb{C}$ be the field of complex numbers. Find the vectors in $C^3$ spanned by the vectors (1, 0, – 1), (0, 1, 1) and (1, 1, 1).
2. Let 0 be the zero vector of a vector space V. Prove that $v0 = 0$ for all $v \in V$.
3. Let S be a non-empty subset S of a vector space V. Prove that S is a subspace of V if $\alpha u + \beta v \in S$ for every $u, v \in S$ and for every scalar $\alpha$, $\beta$.
4. Check whether $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined as $T(x, y) = (x - y, 0)$ is a linear transformation or not.
5. In the Euclidean space $\mathbb{R}^3$ with standard innerproduct, prove that the vectors (1, 0, 1), (– 1, 0, 1) and (0, 1, 0) is an orthogonal set of vectors.
6. Prove that the similar matrices have same characteristic polynomials.
7. Let $W_1$, ..., $W_2$ be subspaces of a vector space V. Prove that $W_1$ and $W_2$ are linearly independent if and only if $W_1 \cap W_2 = \{0\}$.
8. Let $\alpha = (x_1, x_2)$ and $\beta = (y_1, y_2)$ be element in $\mathbb{R}^2$. Prove that $(\alpha \mid \beta) = x_1y_1 - x_2y_1 - x_1y_2 + 4x_2y_2$ is an inner product on $\mathbb{R}^2$.

(8 × 1 = 8 weightage)

## Part B

*Answer any* **six** *by choosing* **two** *questions from each unit.*
*Each question carries* 2 *weightage.*

Unit I

9. Let $\mathbb{F}$ be a field. prove that $\mathbb{F}^n$ is a vector space.
10. Let S be a subset of a vector space V. Prove that a subspace spanned by S is the set of all linear combinations of elements in S.

**Turn over**

11. If A is an $n \times n$ matrix over $\mathbb{F}$ and B, C are $n \times p$ matrices over $\mathbb{F}$. Prove that for any scalar $k$, $A(kB + C) = kAB + AC$.

Unit II

12. Prove or disprove: For any subset S of a finite dimensional vector space V, $(S^\circ)^\circ$ is the subspace spanned by S.

13. Let $T : \mathbb{C}^2 \to \mathbb{C}^2$ defined as projection to the first component. Let $\mathcal{B}$ be the standard basis for $\mathbb{C}^2$ and let $\mathcal{B}' = \{(1, i), (-i, 2)\}$. Find the matrix relative to the pair, $\mathcal{B}$, $\mathcal{B}'$.

14. Let $W_1$ and $W_2$ be subspaces of a finite dimensional vector space. Prove that $W_1 = W_2$ if and only $W_1^\circ = W_2^\circ$.

Unit III

15. Prove that an orthogonal set of non-zero vectors is linearly independent.

16. Let V be a finite dimensional vector space. Let $W_1, \ldots, W_k$ be subspaces of V and let $\mathcal{B}_i$ be ordered basis for each $W_i$ for $i = 1, \ldots, k$. If $W_1, \ldots W_k$ is linearly independent prove that the sequence $\mathcal{B} = \{\mathcal{B}_1, \ldots, \mathcal{B}_k\}$ is an ordered basis for $W_1 + \ldots + W_k$.

17. If $W_1$ be any subspace of a finite dimensional vector space V then prove the existence of a subspace $W_2$ of V such that $V = W_1 \oplus W_2$.

(6 × 2 = 12 weightage)

## Part C

*Answer any* **two** *questions.*
*Each question carries* 5 *weightage.*

18. Let V be a vector space which is spanned by a finite set of vectors $v_1, \ldots, u_m$. Prove that any independent set of vectors in V is finite and contains no more than $m$ elements.

19. Let V be a vector space over a field F and let $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ be an ordered basis for V. Let W be a vector space over the same field and let $\beta_1, \beta_2, \ldots, \beta_n$ be any $n$ vectors in W. Prove that these exist a unique linear transformation $T : V \to W$ such that $T_{\alpha_i} = \beta_i$ for all $i = 1, 2, \ldots, n$.

20. Let $\theta$ be a real number. Prove that the following two matrices are similar over $\mathbb{C}$,

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, B = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}.$$

21. Let T be a linear operator on a finite dimensional vector space V, Prove that the minimal polynomial for T divides its characteristic polynomial. Let T be a linear operator on a vector space $V = W_1 \oplus \ldots \oplus W_k$. Then prove that there exist projections $E_i$ on V such that

(a) $T = c_1E_1 + \ldots + c_kE_k$.

(b) $I = E_1 + \ldots + E_k$.

(c) $E_iE_j = 0, i \neq j$ and

(d) The range of $E_i$ is the characteristic space for T that is associated with $c_i$.

(2 × 5 = 10 weightage)